

Simple harmonic motion with damping

Key Points

Damping means there is some force dependent on velocity (linear damping) in SHM so

$$\ddot{x} + k\dot{x} + \omega^2 x = 0$$

With damping types being determined by auxiliary equation:

- Light damping if $k^2 < 4\omega^2$ and roots $\text{Re}(m) < 0$ ($k > 0$)
- Heaving damping if $k^2 > 4\omega^2$ and roots $m < 0$ ($k > 0$)
- Critical damping if $k^2 = 4\omega^2$ and root $m < 0$ ($k > 0$)

Basic Skills

Q1. For simple harmonic motion (without damping)

- a) State the differential equation
- b) State the solution for simple harmonic motion

Q2. Find the equation for displacement (from equilibrium position) for a spring-mass oscillator with mass 3kg and stiffness constant 30Nm^{-1} initially displaced 30cm from equilibrium and released from rest

Q3. For the three types of damping; draw a graph representing that type and explain what the main feature of that damping compared to the others is

Competency Skills

Q4. For the differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$ state how the discriminant of the auxiliary equation determines the types of solutions, pointing out the requirements such that the solution tends to zero as we go to infinity

Q5. Solved the damped oscillations modelled by $\ddot{x} + k\dot{x} + 4x = 0$ with $x(0) = 0, \dot{x}(0) = 1$ for

- a) $k = 5$ b) $k = 2$ c) $k = 4$

Q6. Explain why the differential equation $\ddot{x} - 5\dot{x} + 4x = 0$ is not a damped oscillation

Q7. A spring-mass oscillator has mass 2kg and stiffness constant 50Nm^{-1} initially displaced 10cm from equilibrium and released from rest and it is linearly damped. Determine the damping constant such that it is critically damped

Advanced Skills

Q8. A block of mass 40kg is dragged along a horizontal surface by a spring which has a natural length of 0.4m with the end of the spring not attached to the block being pulled at a constant velocity of 2ms^{-1} . The block is subject to a resistive force equal to twice the velocity v of the block and using $l(t)$ as the length of the string at time t

- a) Show that resultant force on the block is $F = k(l - 0.4) - 2v$
- b) Explain why $\frac{dl}{dt} = 2 - v$
- c) Hence show that $\frac{d^2l}{dt^2} + 0.05\frac{dl}{dt} + \frac{k}{40}l = 0.1 + 0.01k$
- d) Hence find the general solution for $l(t)$ when $k = \frac{4}{250}$

Extension

Q9. Research the dashpot force and write an explanation of how it damps simple harmonic motion

Hint: Consider the possible solutions to the ODE